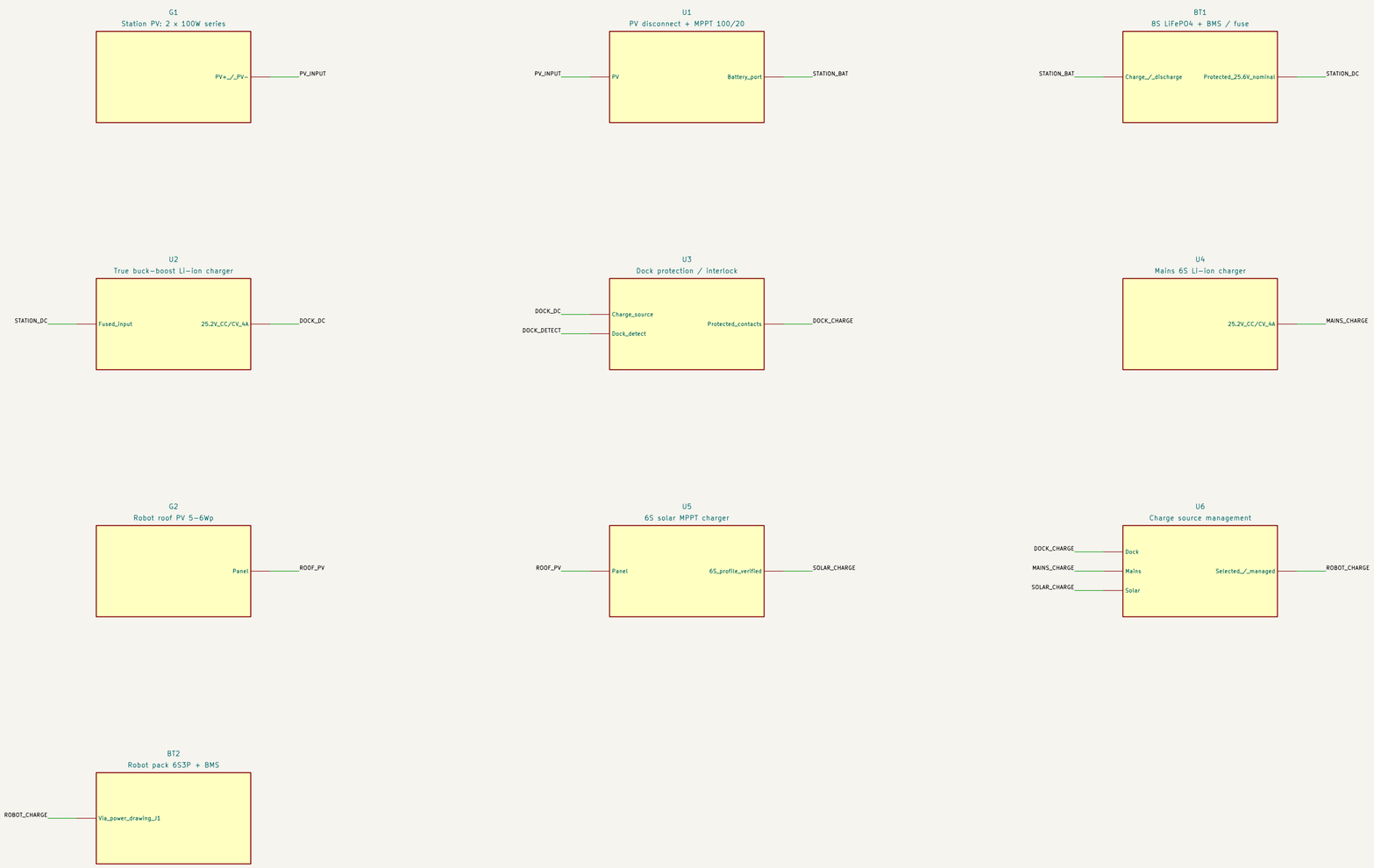


Cable-level diagram: every power interface includes positive AND protected return. No connection to raw cell B-.

Station LiFePO4 and robot Li-ion need DIFFERENT charging profiles. Do not tie the batteries together.

Use a true buck-boost charger across the station battery voltage range. CC/CV supply alone is not charge management.

Charge-source management must prevent backfeed, enforce combined current/temperature limits and preserve termination.

Simplest prototype: one charge source selected at a time, robot off while charging, appropriate BMS connected.

Dock needs current limiting, contact sequencing, polarity protection and motion inhibit; Hall detection alone is insufficient.

Validate series-panel cold Voc against 100V MPPT limit and Vmp headroom. Roof solar is supplemental, not BA charging.

Optional independent beacon is not included in the shared robot rail; give it its own protected battery/charger power path.

Symbol terminal numbers are logical IDs, NOT connector pin numbers
MODULE INTERCONNECTIONS ONLY — NOT FOR PCB FABRICATION

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File: underbot_charging.kicad_sch

Title: UnderBot charging architecture — separate robot and station batteries

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